Unit A

Multiple Choice Bank — 75 Questions

Question Set

MC 1. According to the molecular explanation in the lesson, why does heat transfer occur from hot to cold substances?

A. Hot molecules are larger and push cold molecules away
B. Fast-moving molecules collide with slower molecules, transferring kinetic energy
C. Cold molecules attract heat energy through electromagnetic forces
D. Hot molecules break apart and release energy particles

Correct answer: B

Rationale: Heat transfer occurs because molecules in hot substances move faster and collide with slower-moving molecules in cooler substances, transferring kinetic energy through these molecular collisions until thermal equilibrium is reached.

MC 2. What mass of water can be heated from 20.0°C to 85.0°C using 67,990 J of energy? (c_{water} = 4.184 J/g°C)

A. 125.0 g
B. 500.0 g
C. 250.0 g
D. 375.0 g

Correct answer: C

Rationale: Using $Q = mc\Delta t$: 67,990 J = m × 4.184 J/g°C × (85.0°C - 20.0°C) = m × 4.184 × 65.0. Solving for m gives 250.0 g, as shown in the lesson example.

MC 3. In a calorimetry experiment, 50.0 g of hot metal at 95.0°C is placed in 200.0 g of water at 22.0°C, reaching a final temperature of 28.5°C. What is the specific heat capacity of the metal?

A. 0.82 J/g°C
B. 3.28 J/g°C
C. 2.46 J/g°C
D. 1.64 J/g°C

Correct answer: D

Rationale: Using $Q_{lost} = Q_{gained}$ and the calculation shown in the lesson: (50.0 g)(c_{metal})(66.5°C) = (200.0 g)(4.184 J/g°C)(6.5°C), solving gives c_{metal} = 1.64 J/g°C.

MC 4. Why is the specific heat capacity of water relatively high compared to most substances?

A. Water molecules form hydrogen bonds that require additional energy to overcome during heating
B. Water has a low molecular mass allowing rapid energy absorption
C. Water molecules are polar and repel heat energy
D. Water contains dissolved minerals that store extra heat

Correct answer: A

Rationale: The lesson states that water's high specific heat capacity is due to hydrogen bonds between water molecules that require additional energy to overcome during heating, making water an excellent thermal reservoir.

MC 5. What fundamental principle allows us to set up equations where $Q_{lost} = Q_{gained}$ in calorimetry?

A. The second law of thermodynamics
B. Hess's law of constant heat summation

C. Energy conservation - energy cannot be created or destroyed
D. The principle of thermal equilibrium

Correct answer: C

Rationale: The lesson emphasizes that energy conservation is the fundamental principle: energy cannot be created or destroyed, only transferred, so heat lost by hot objects equals heat gained by cold objects.

MC 6. At thermal equilibrium, what characteristic do all molecules in the system share?

A. The same mass
B. The same average kinetic energy
C. The same specific heat capacity
D. The same total heat content

Correct answer: B

Rationale: At thermal equilibrium, all molecules have the same average kinetic energy and therefore the same temperature, as stated in the lesson's explanation of heat transfer.

MC 7. When 250.0 g of water is heated from 20.0°C to 85.0°C, what happens at the molecular level?

A. Hydrogen bonds between water molecules break permanently
B. Water molecules undergo a phase change to steam
C. Molecular motion increases without breaking hydrogen bonds
D. New chemical bonds form between water molecules

Correct answer: C

Rationale: The lesson explains that absorbed energy increases vibrational and rotational motion of water molecules within the liquid phase, without breaking hydrogen bonds (which would cause a phase change).

MC 8. Why does heat flow spontaneously from hot to cold according to thermodynamics?

A. It decreases the total energy of the system
B. It increases the entropy (disorder) of the universe
C. It creates new energy in the cold object
D. It reduces molecular collisions

Correct answer: B

Rationale: The lesson explains that heat flows spontaneously from hot to cold because this increases total entropy, making energy more evenly distributed and increasing overall disorder, as required by the second law of thermodynamics.

MC 9. What is the distinction between temperature and heat as described in the lesson?

A. Temperature is measured in joules, heat in degrees Celsius
B. Temperature is total energy, heat is average energy
C. Temperature is average kinetic energy, heat is total kinetic energy transfer
D. Temperature and heat are identical concepts

Correct answer: C

Rationale: The lesson clearly distinguishes that temperature represents average kinetic energy of molecules, while heat represents the total kinetic energy transfer between substances.

MC 10. In the equation $Q = mc\Delta t$, what does Δt specifically represent?

A. Initial temperature minus final temperature
B. Final temperature minus initial temperature
C. Average temperature of the process
D. Absolute temperature in Kelvin

Correct answer: B

Rationale: The lesson defines Δt as the temperature change calculated as final temperature minus initial temperature.

MC 11. According to the lesson, what type of calorimeter is used to measure caloric content of foods?

A. Coffee cup calorimeter
B. Bomb calorimeter
C. Ice calorimeter
D. Solution calorimeter

Correct answer: B

Rationale: The lesson states that food scientists employ bomb calorimeters to measure the caloric content of foods in controlled combustion environments.

MC 12. How do beverage companies apply calorimetry principles according to the lesson?

A. To measure carbonation levels
B. To control fermentation temperatures
C. To determine the energy content of drinks
D. To test packaging insulation

Correct answer: C

Rationale: The lesson mentions that beverage companies use calorimetry to determine the energy content of drinks.

MC 13. What industry uses calorimetry to determine specific heat capacities of alloys?

A. Pharmaceutical industry
B. Metallurgical industry
C. Textile industry
D. Agricultural industry

Correct answer: B

Rationale: The lesson specifically states that metallurgical industries use calorimetry to determine specific heat capacities of alloys.

MC 14. When calculating heat transfer for water, what is the specific heat capacity value used in the lesson?

A. 1.00 J/g°C
B. 2.09 J/g°C
C. 4.184 J/g°C
D. 8.37 J/g°C

Correct answer: C

Rationale: The lesson consistently uses 4.184 J/g°C as the specific heat capacity of water in all calculations.

MC 15. In an isolated system during heat transfer, what remains constant?

A. Temperature of each substance
B. Specific heat capacity
C. Total energy
D. Rate of heat transfer

Correct answer: C

Rationale: The lesson states that in any isolated system, the total energy remains constant due to energy conservation.

MC 16. What unit is used for heat energy (Q) in the $Q = mc\Delta t$ equation?

A. Grams

B. Degrees Celsius
C. Joules
D. Watts

Correct answer: C

Rationale: The lesson specifies that Q represents heat energy transferred, measured in joules.

MC 17. According to the biological rule stated, where does all chemical energy in biological systems originate?

A. Geothermal vents
B. Chemical reactions in soil
C. Radioactive decay
D. Solar energy captured through photosynthesis

Correct answer: D

Rationale: The Big Biological Rule states that all chemical energy in biological systems originates from solar energy captured through photosynthesis.

MC 18. Why do hydrocarbons have higher energy density than alcohols according to the structure-function relationship?

A. C-H bonds store more energy than C-O bonds
B. Hydrocarbons have lower molecular mass
C. Alcohols contain more hydrogen atoms
D. Hydrocarbons are more polar molecules

Correct answer: A

Rationale: The Structure → Function Reminder explains that C-H bonds store more energy than C-O bonds, making hydrocarbons more energy-dense than alcohols.

MC 19. What does the lesson indicate about catalysts and enthalpy changes?

A. Catalysts increase the enthalpy change of reactions
B. Catalysts decrease the enthalpy change of reactions
C. Catalysts never change enthalpy changes
D. Catalysts only affect endothermic reactions

Correct answer: C

Rationale: The Final Diploma Alert states that catalysts affect reaction rates but never change enthalpy changes or equilibrium positions.

MC 20. What distinction does the diploma exam frequently test according to the lesson?

A. Between activation energy and enthalpy change
B. Between mass and weight
C. Between temperature and pressure
D. Between catalysts and inhibitors

Correct answer: A

Rationale: The Final Diploma Alert specifically mentions that diploma exams frequently test the distinction between activation energy and enthalpy change.

MC 21. According to the lesson, what determines the energy content of molecules?

A. Their molecular mass
B. Their bond structure
C. Their physical state only
D. Their color and appearance

Correct answer: B

Rationale: The lesson states that energy content of molecules depends directly on their bond structure, with different bonds storing different amounts of energy.

MC 22. What process creates more energy-dense fossil fuels according to the lesson?

A. Adding hydrogen atoms
B. Increasing pressure only
C. Removing oxygen atoms from biological matter
D. Adding nitrogen compounds

Correct answer: C

Rationale: The lesson explains that geological processes that remove oxygen atoms from biological matter create more energy-dense fossil fuels.

MC 23. In the cause-effect chain presented, what immediately follows 'solar photons'?

A. Geological transformation
B. Combustion
C. Photosynthesis
D. Heat release

Correct answer: C

Rationale: The Cause → Effect Summary shows the sequence: Solar photons → photosynthesis → chemical bond formation...

MC 24. What must be specified in thermochemical equations according to the lesson?

A. The laboratory temperature
B. The catalyst used
C. Physical states of substances
D. The reaction vessel material

Correct answer: C

Rationale: The lesson states that physical states must be specified in thermochemical equations because intermolecular forces affect phase-dependent enthalpy values.

MC 25. According to the lesson, what enables quantitative energy predictions in chemistry?

A. Standardized formation enthalpies and Hess's law
B. Temperature measurements alone
C. Catalyst selection
D. Pressure variations

Correct answer: A

Rationale: The Master Takeaways state that thermochemical calculations enable quantitative energy predictions through standardized formation enthalpies and Hess's law.

MC 26. What does activation energy affect according to the lesson?

A. Total energy released only
B. Reaction rates and initiation requirements
C. Equilibrium position only
D. Product stability

Correct answer: B

Rationale: The lesson distinguishes that activation energy affects reaction rates and initiation requirements, while enthalpy change determines total energy released or absorbed.

MC 27. How does calorimetry bridge theoretical calculations with experimental reality?

A. By eliminating all experimental error

B. By using $Q = mc\Delta t$ to measure energy changes

C. By avoiding the need for calculations

D. By using computer simulations only

Correct answer: B

Rationale: The Master Takeaways explain that calorimetry bridges theory with reality by using $Q = mc\Delta t$ to measure energy changes that validate predicted values.

MC 28. What does the lesson say about energy released during combustion compared to energy captured from sunlight?

A. Combustion releases more energy than was captured

B. Combustion releases less energy than was captured

C. Combustion releases exactly the amount originally captured

D. The amounts are unrelated

Correct answer: C

Rationale: The Big Biological Rule states that energy conservation requires combustion to release exactly the amount of energy originally captured from sunlight.

MC 29. What industrial application is mentioned in the final transfer prompt?

A. Converting water to hydrogen

B. Converting methane to ethane

C. Converting ethanol to acetone

D. Converting methanol to formaldehyde

Correct answer: D

Rationale: The Final Transfer Prompt specifically mentions designing an energy analysis for converting methanol (CH_3OH) to formaldehyde (CH_2O) plus hydrogen gas.

MC 30. What happens to energy distribution when heat flows from hot to cold?

A. Energy becomes more concentrated

B. Energy becomes more evenly distributed

C. Energy disappears from the system

D. Energy converts to matter

Correct answer: B

Rationale: The lesson explains that when heat flows from hot to cold, energy becomes more evenly distributed, increasing the overall disorder of the system.

MC 31. What type of molecular motion increases when water is heated without phase change?

A. Nuclear motion only

B. Translational motion only

C. Vibrational and rotational motion

D. Electron orbital transitions

Correct answer: C

Rationale: The lesson specifies that heating water involves energy absorption through increased vibrational and rotational motion of molecules.

MC 32. According to the lesson, what must be included in the industrial process energy analysis?

A. Only the enthalpy change

B. Only catalyst requirements

C. Predicted enthalpy change, heating requirements, catalyst needs, and safety considerations

D. Only safety considerations

Correct answer: C

Rationale: The Final Transfer Prompt requires analysis including predicted enthalpy change, whether external heating is needed, catalyst requirements, and safety considerations based on energy calculations.

MC 33. What fundamental concept connects all thermochemical phenomena according to the lesson?

A. Temperature gradients
B. Pressure differences
C. Energy conservation
D. Molecular size

Correct answer: C

Rationale: The Master Takeaways state that energy conservation connects all thermochemical phenomena, including heat transfer, bond energy, and enthalpy changes.

MC 34. Why must ΔH values be properly scaled with equation coefficients?

A. This is a frequent diploma exam topic
B. To change the activation energy
C. To alter the equilibrium position
D. To modify catalyst effectiveness

Correct answer: A

Rationale: The Final Diploma Alert mentions that proper scaling of ΔH values with equation coefficients is frequently tested on diploma exams.

MC 35. What explains why a large container of lukewarm water can contain more heat than a small container of hot water?

A. Lukewarm water has higher specific heat
B. Temperature is average kinetic energy while heat is total energy transfer
C. Hot water loses energy faster
D. Container size affects specific heat capacity

Correct answer: B

Rationale: The lesson explains this by distinguishing temperature (average kinetic energy) from heat (total kinetic energy transfer), allowing larger masses to contain more total heat despite lower temperature.

MC 36. What role do intermolecular forces play in thermochemical equations?

A. They determine reaction rate only
B. They affect phase-dependent enthalpy values
C. They prevent all chemical reactions
D. They only matter in gases

Correct answer: B

Rationale: The Master Takeaways note that intermolecular forces affect phase-dependent enthalpy values, explaining why physical states must be specified in thermochemical equations.

MC 37. What does the lesson say about Hess's law application?

A. It only works for exothermic reactions
B. It must be correctly applied on diploma exams
C. It changes with temperature
D. It requires a catalyst

Correct answer: B

Rationale: The Final Diploma Alert states that the correct application of Hess's law is tested on diploma exams.

MC 38. What determines thermodynamic favorability according to the lesson?

A. Activation energy only
B. Catalyst presence
C. Enthalpy change
D. Reaction vessel size

Correct answer: C

Rationale: The lesson distinguishes that enthalpy change determines thermodynamic favorability, while activation energy affects reaction rates.

MC 39. According to the lesson, what is the relationship between geological time scales and energy conservation?

A. Energy decreases over geological time
B. Energy increases during fossilization
C. Energy is conserved regardless of time scales
D. Time affects the amount of energy stored

Correct answer: C

Rationale: The Big Biological Rule emphasizes that energy conservation applies regardless of geological time scales involved in the transformation from photosynthesis to fossil fuels.

MC 40. What foundation does thermochemical calculation provide according to the lesson?

A. Only academic understanding
B. Industrial process design and fuel efficiency analysis
C. Temperature measurement techniques
D. Catalyst manufacturing methods

Correct answer: B

Rationale: The Master Takeaways state that thermochemical calculations provide the foundation for industrial process design and fuel efficiency analysis.

MC 41. What experimental tool validates predicted thermochemical values?

A. Spectroscopy
B. Chromatography
C. Calorimetry
D. Microscopy

Correct answer: C

Rationale: The Master Takeaways explain that calorimetry validates predicted values and enables determination of unknown thermochemical parameters.

MC 42. In the energy transformation sequence, what directly follows 'fossil fuel creation'?

A. Photosynthesis
B. Geological transformation
C. Combustion
D. Solar photons

Correct answer: C

Rationale: The Cause → Effect Summary shows: fossil fuel creation → combustion → heat release...

MC 43. What must reactions predicted by thermochemical calculations sometimes require to proceed?

A. A catalyst only
B. Lower pressure
C. External energy input
D. Removal of products

Correct answer: C

Rationale: The lesson notes that even highly exothermic reactions may require external energy input to proceed, relating to activation energy concepts.

MC 44. How are energy storage in bonds and energy release during combustion related?

A. They are unrelated processes
B. Energy stored in bonds equals energy released during combustion
C. Combustion creates more energy than stored
D. Bond energy is always greater than combustion energy

Correct answer: B

Rationale: The Master Takeaways state that energy stored in bonds equals energy released during combustion as part of energy conservation.

MC 45. What aspect of reactions do catalysts affect according to the diploma alert?

A. Equilibrium positions
B. Enthalpy changes
C. Reaction rates
D. Product composition

Correct answer: C

Rationale: The Final Diploma Alert states that catalysts affect reaction rates but never change enthalpy changes or equilibrium positions.

MC 46. What makes the $Q = mc\Delta t$ equation useful in chemistry?

A. It eliminates the need for experiments
B. It relates four measurable quantities
C. It only works for water
D. It requires no unit analysis

Correct answer: B

Rationale: The lesson states that $Q = mc\Delta t$ quantifies heat transfer by relating four measurable quantities: heat energy, mass, specific heat capacity, and temperature change.

MC 47. What type of energy analysis does the final transfer prompt require for safety considerations?

A. Pressure buildup only
B. Temperature monitoring only
C. Energy release calculations
D. Catalyst toxicity only

Correct answer: C

Rationale: The Final Transfer Prompt specifically mentions safety considerations based on energy release calculations.

MC 48. According to the lesson, what creates predictable mathematical relationships in heat transfer?

A. Random molecular motion
B. Each variable in $Q = mc\Delta t$ directly affecting heat transferred
C. Catalyst presence
D. Container material

Correct answer: B

Rationale: The lesson states that each variable in $Q = mc\Delta t$ directly affects the amount of heat transferred, creating predictable mathematical relationships.

MC 49. What enables determination of unknown thermochemical parameters?

A. Theoretical calculations only
B. Computer simulations
C. Calorimetry measurements using $Q = mc\Delta t$
D. Visual observation

Correct answer: C

Rationale: The Master Takeaways state that calorimetry using $Q = mc\Delta t$ enables determination of unknown thermochemical parameters.

MC 50. What concept is described as independent of reaction pathway?

A. Activation energy
B. Reaction rate
C. Enthalpy changes
D. Catalyst effectiveness

Correct answer: C

Rationale: The Master Takeaways state that enthalpy changes are independent of reaction pathway, a key principle of Hess's law.

MC 51. In the calorimetry example, what was the temperature change of the water when the metal was added?

A. 66.5°C
B. 6.5°C
C. 28.5°C
D. 73.0°C

Correct answer: B

Rationale: In the example, water temperature changed from 22.0°C to 28.5°C, giving Δt = 28.5°C - 22.0°C = 6.5°C.

MC 52. What is required to overcome during water heating that makes its specific heat capacity high?

A. Covalent bonds within molecules
B. Hydrogen bonds between molecules
C. Ionic attractions
D. Van der Waals forces only

Correct answer: B

Rationale: The lesson explains that water's high specific heat capacity results from hydrogen bonds between molecules that require additional energy to overcome during heating.

MC 53. According to the lesson, what allows calculation of energy in industrial processes?

A. Hess's law and formation enthalpies
B. Temperature measurements only
C. Pressure readings
D. Visual color changes

Correct answer: A

Rationale: The lesson indicates that standardized formation enthalpies and Hess's law enable quantitative energy predictions for industrial process design.

MC 54. What molecular property makes water an excellent thermal reservoir?

A. Low molecular mass
B. High specific heat capacity due to hydrogen bonding
C. Ionic character
D. Metallic bonding

Correct answer: B

Rationale: The lesson states that water's high specific heat capacity, resulting from hydrogen bonding, makes it an excellent thermal reservoir.

MC 55. In calorimetry calculations, what equation relationship is used when hot metal is placed in water?

A. Q_metal = Q_water
B. Q_metal = -Q_water
C. Q_metal + Q_water = 100
D. Q_metal × Q_water = constant

Correct answer: B

Rationale: The lesson shows that Q_metal = -Q_water because heat lost by the metal equals heat gained by the water (with opposite signs).

MC 56. What determines whether a reaction will require external heating according to the industrial analysis?

A. The catalyst used
B. The predicted enthalpy change
C. The container size
D. The time of day

Correct answer: B

Rationale: The Final Transfer Prompt indicates that predicted enthalpy change determines whether external heating will be required for the process.

MC 57. What is the temperature change when 50.0 g of metal at 95.0°C reaches equilibrium at 28.5°C?

A. 28.5°C
B. 95.0°C
C. -66.5°C
D. 123.5°C

Correct answer: C

Rationale: Δt = final - initial = 28.5°C - 95.0°C = -66.5°C, showing the metal cooled by 66.5°C.

MC 58. According to the lesson, what type of collisions transfer energy during heat flow?

A. Nuclear collisions
B. Molecular collisions between fast and slow molecules
C. Electron-electron collisions only
D. Photon absorption

Correct answer: B

Rationale: The lesson explains that faster-moving molecules collide with slower-moving molecules, transferring kinetic energy through these molecular collisions.

MC 59. What happens to hydrogen bonds when water is heated within the liquid phase?

A. They break permanently
B. They remain intact while molecular motion increases
C. They become stronger
D. They convert to covalent bonds

Correct answer: B

Rationale: The lesson clarifies that absorbed energy does not break hydrogen bonds (which would cause phase change) but increases molecular motion within the liquid phase.

MC 60. What calculation gives 5439.2 J in the metal calorimetry example?

A. (50.0 g)(1.64 J/g°C)(66.5°C)
B. (200.0 g)(4.184 J/g°C)(6.5°C)
C. (250.0 g)(4.184 J/g°C)(20.0°C)
D. (100.0 g)(2.09 J/g°C)(26.0°C)

Correct answer: B

Rationale: The calculation shows Q_water = (200.0 g)(4.184 J/g°C)(6.5°C) = 5439.2 J for the heat gained by water.

MC 61. According to the lesson, what must students demonstrate understanding of regarding catalysts?

A. Catalysts change equilibrium positions
B. Catalysts affect rates but not enthalpy changes or equilibrium
C. Catalysts increase product yield
D. Catalysts reduce activation energy to zero

Correct answer: B

Rationale: The Final Diploma Alert emphasizes students must understand that catalysts affect reaction rates but never change enthalpy changes or equilibrium positions.

MC 62. What is the result of converting 67,990 J to kilojoules?

A. 6.80 kJ
B. 680 kJ
C. 68.0 kJ
D. 0.068 kJ

Correct answer: C

Rationale: The lesson shows the conversion: 67,990 J = 68.0 kJ (dividing by 1000 to convert J to kJ).

MC 63. What sequence correctly represents the energy flow from sun to industrial use?

A. Combustion → photosynthesis → fossil fuels → heat
B. Solar photons → photosynthesis → fossil fuels → combustion
C. Fossil fuels → solar energy → combustion → industry
D. Industry → combustion → photosynthesis → solar energy

Correct answer: B

Rationale: The Cause → Effect Summary shows: Solar photons → photosynthesis → geological transformation → fossil fuel creation → combustion.

MC 64. What property of molecules determines their average kinetic energy?

A. Mass
B. Temperature
C. Pressure
D. Volume

Correct answer: B

Rationale: The lesson states that temperature represents the average kinetic energy of molecules.

MC 65. How much energy is transferred when 250.0 g of water undergoes a 65.0°C temperature change?

A. 34.0 kJ
B. 136 kJ
C. 68.0 kJ
D. 17.0 kJ

Correct answer: C

Rationale: Q = (250.0 g)(4.184 J/g°C)(65.0°C) = 67,990 J = 68.0 kJ as shown in the lesson example.

MC 66. What industrial consideration is mentioned for the methanol to formaldehyde conversion?

A. Color change monitoring
B. Safety considerations based on energy release
C. Product taste testing
D. Magnetic field effects

Correct answer: B

Rationale: The Final Transfer Prompt requires safety considerations based on energy release calculations for the industrial process.

MC 67. What happens to molecular speed when a substance is heated?

A. Molecules slow down
B. Molecules move faster
C. Molecular speed remains constant
D. Some molecules stop moving

Correct answer: B

Rationale: The lesson explains that heating increases average kinetic energy, causing molecules to move faster.

MC 68. What type of bond has higher energy content according to the structure-function relationship?

A. C-O bonds
B. C-C bonds only
C. C-H bonds
D. O-H bonds only

Correct answer: C

Rationale: The Structure → Function Reminder specifically states that C-H bonds store more energy than C-O bonds.

MC 69. In the equation (3325 g°C)(c_metal) = 5439.2 J, what value of c_metal is calculated?

A. 0.61 J/g°C
B. 1.64 J/g°C
C. 2.46 J/g°C
D. 3.28 J/g°C

Correct answer: B

Rationale: Solving: c_metal = 5439.2 J ÷ 3325 g°C = 1.64 J/g°C as shown in the lesson calculation.

MC 70. What fundamental principle states that entropy of the universe must increase?

A. First law of thermodynamics
B. Second law of thermodynamics
C. Third law of thermodynamics
D. Zeroth law of thermodynamics

Correct answer: B

Rationale: The lesson identifies the second law of thermodynamics as stating that entropy (disorder) of the universe must increase.

MC 71. What geological process creates energy-dense fossil fuels from biological matter?

A. Adding water molecules
B. Increasing oxygen content
C. Removing oxygen atoms

D. Adding nitrogen compounds

Correct answer: C

Rationale: The lesson explains that geological processes removing oxygen atoms from biological matter create more energy-dense fossil fuels.

MC 72. According to the lesson, what concept explains why reactions may need energy input despite being exothermic?

A. Enthalpy change
B. Activation energy
C. Equilibrium constant
D. Rate constant

Correct answer: B

Rationale: The lesson notes that activation energy concepts explain why even highly exothermic reactions may require external energy input to proceed.

MC 73. What happens at the point of thermal equilibrium between two substances?

A. All molecular motion stops
B. Energy transfer continues in one direction
C. Both substances have the same temperature
D. The substances chemically react

Correct answer: C

Rationale: At thermal equilibrium, all molecules have the same average kinetic energy and therefore the same temperature.

MC 74. What relationship does Hess's law establish for enthalpy changes?

A. They depend on the catalyst used
B. They are independent of reaction pathway
C. They increase with temperature
D. They decrease over time

Correct answer: B

Rationale: The lesson emphasizes that enthalpy changes are independent of reaction pathway, which is the principle behind Hess's law.

MC 75. What final product is mentioned in the Cause → Effect chain after calorimetric measurement?

A. Solar panel design
B. Optimized chemical processes
C. New catalyst development
D. Temperature control systems

Correct answer: B

Rationale: The Cause → Effect Summary ends with: calorimetric measurement → enthalpy determination → thermochemical calculations → industrial energy planning → optimized chemical processes.

Unit A

Constructed Response Bank

Question 1

A student claims, 'Heat always flows from substances with more total heat energy to substances with less total heat energy.'

a. Explain why this claim is incorrect by distinguishing between temperature and heat.

b. Describe what happens at the molecular level when heat transfers between substances.

c. State one industrial application of calorimetry mentioned in the lesson.

Question 2

An experiment shows that 250.0 g of water requires 67,990 J to heat from 20.0°C to 85.0°C.

a. Calculate the specific heat capacity of water using the given data.

b. Explain why water has a relatively high specific heat capacity compared to most substances.

c. State what happens to the hydrogen bonds when water is heated within the liquid phase.

Question 3

A researcher states, 'The second law of thermodynamics explains why heat flows spontaneously from hot to cold objects.'

a. Explain how the second law of thermodynamics relates to spontaneous heat flow.

b. Describe what thermal equilibrium means in terms of molecular kinetic energy.

c. State the fundamental principle that allows us to write $Q_{lost} = Q_{gained}$ in calorimetry.

Question 4

In a calorimetry experiment, 50.0 g of hot metal at 95.0°C is placed in 200.0 g of water at 22.0°C, and the final temperature reaches 28.5°C.

a. Calculate the heat gained by the water in this experiment.

b. Calculate the specific heat capacity of the metal using energy conservation principles.

c. State which industry uses calorimetry to determine specific heat capacities of materials like this metal.

Question 5

A chemistry teacher says, 'All chemical energy in biological systems can be traced back to the sun.'

a. Explain the energy transformation pathway from solar energy to fossil fuels.

b. Explain why hydrocarbons have higher energy density than alcohols based on bond structure.

c. State what type of calorimeter food scientists use to measure caloric content.

Question 6

Research shows that catalysts can change reaction rates but have specific limitations in chemical processes.

a. Explain what aspects of a chemical reaction catalysts can and cannot change.

b. Explain the difference between activation energy and enthalpy change in determining reaction characteristics.

c. State why even highly exothermic reactions may require external energy input.

Question 7

An industrial process converts methanol (CH_3OH) to formaldehyde (CH_2O) plus hydrogen gas.

a. List the four components that must be included in an energy analysis of this industrial process.

b. Explain how thermochemical calculations provide the foundation for industrial applications.

c. State what principle allows enthalpy calculations to be independent of reaction pathway.

Question 8

A student observes that heating 250.0 g of water by 65.0°C requires 68.0 kJ of energy.

a. Show the complete calculation that confirms this energy requirement using $Q = mc\Delta t$.

b. Explain what happens to water molecules when this energy is absorbed without phase change.

c. State what each variable in the equation $Q = mc\Delta t$ represents with its units.

Question 9

Researchers find that molecular structure directly determines the energy content of fuel molecules.

a. Explain the relationship between bond types and energy content in organic molecules.

b. Explain how geological processes create energy-dense fossil fuels from biological matter.

c. State why physical states must be specified in thermochemical equations.

Question 10

A coffee cup calorimeter experiment demonstrates fundamental thermochemical principles in the laboratory.

a. Explain how calorimetry bridges theoretical calculations with experimental reality.

b. Explain why the equation Q_lost = Q_gained can be used in calorimetry calculations.

c. State three industries mentioned that use calorimetry applications.

Question 11

A diploma exam question asks students to distinguish between fundamental thermochemical concepts.

a. Explain three key distinctions that diploma exams frequently test according to the lesson.

b. Explain why enthalpy changes are independent of reaction pathway while activation energy is not.

c. State what students must understand about catalysts' effects on equilibrium.

Question 12

An experiment demonstrates that large amounts of lukewarm water can contain more total heat than small amounts of very hot water.

a. Explain this observation using the concepts of temperature and heat.

b. Predict the direction of heat flow if these two water samples are brought into contact.

c. State what determines the direction of spontaneous heat flow.

Question 13

The transformation of solar energy to usable chemical energy follows a specific sequence according to the lesson material.

a. List the complete cause-effect sequence from solar photons to optimized chemical processes.

b. Explain how energy conservation applies throughout this entire transformation sequence.

c. State what process captures solar energy and stores it in chemical bonds.

Question 14

A metal sample with unknown specific heat capacity is tested using calorimetry principles.

a. Describe the experimental setup needed to determine the metal's specific heat capacity.

b. Explain why this calculation requires the assumption of an isolated system.

c. Calculate the product (3325 g°C) if solving for a metal's specific heat when heat transfer equals 5439.2 J.

Question 15

Water's unique properties make it valuable for thermal applications in industry and nature.

a. Explain what molecular feature gives water its high specific heat capacity.

b. Explain why water is described as an excellent thermal reservoir.

c. State the specific heat capacity value for water used in calculations.

Question 16

The second law of thermodynamics governs the spontaneous direction of heat transfer in all systems.

a. Define entropy and explain its role in spontaneous heat transfer.

b. Describe what happens to the entropy when fast-moving molecules transfer energy to slow-moving molecules.

c. State which law of thermodynamics requires entropy to increase.

Question 17

Industrial chemists must consider multiple factors when designing energy-efficient processes.

a. Explain how formation enthalpies and Hess's law enable industrial energy planning.

b. Explain why understanding both activation energy and enthalpy change is crucial for industrial processes.

c. State what safety consideration is based on energy release calculations.

Question 18

A calorimetry calculation shows that 50.0 g of metal cooling from 95.0°C to 28.5°C releases 5439.2 J to water.

a. Calculate the temperature change of the metal and explain why it is negative.

b. Set up the equation to solve for the metal's specific heat capacity using the given information.

c. State why the heat lost by metal equals the heat gained by water in magnitude.

Question 19

Fossil fuel formation represents a long-term storage mechanism for solar energy captured by ancient organisms.

a. Explain how the energy in fossil fuels relates to the original solar energy captured.

b. Explain why removing oxygen atoms during fossilization increases energy density.

c. State the biological process that initially captures solar energy.

Question 20

A student must analyze the complete energy transformation from sunlight to industrial chemical processes.

a. Identify four key measurement or calculation steps in the energy transformation sequence after combustion occurs.

b. Explain how calorimetry validates theoretical thermochemical predictions.

c. State the final outcome of the complete cause-effect energy transformation chain.

Model Answers & Rubrics

Question 1

Part a: Explain why this claim is incorrect by distinguishing between temperature and heat.

Model Answer: The claim is incorrect because heat flows based on temperature differences, not total heat content. Temperature represents average kinetic energy of molecules, while heat represents total kinetic energy transfer. Heat flows from higher temperature (higher average kinetic energy) to lower temperature, regardless of total heat content. A large container of lukewarm water can contain more total heat than a small container of hot water, but heat would flow from the hot water to the lukewarm water. (2 marks) (2 marks)

Part b: Describe what happens at the molecular level when heat transfers between substances.

Model Answer: When heat transfers, molecules in the hot substance (moving faster with more kinetic energy) collide with slower-moving molecules in the cooler substance. These molecular collisions transfer kinetic energy from fast-moving to slow-moving molecules. This process continues until thermal equilibrium is reached, where all molecules have the same average kinetic energy and temperature. (2 marks) (2 marks)

Part c: State one industrial application of calorimetry mentioned in the lesson.

Model Answer: Beverage companies use calorimetry to determine the energy content of drinks. (OR: Metallurgical industries use calorimetry to determine specific heat capacities of alloys. OR: Food scientists use bomb calorimeters to measure caloric content of foods.) (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for identifying that heat flows based on temperature not total heat; 1 mark for correctly distinguishing temperature (average KE) from heat (total energy transfer)

Part b (2 marks): 1 mark for describing molecular collisions; 1 mark for explaining energy transfer until equilibrium

Part c (1 mark): 1 mark for correctly stating any industrial application from the lesson

Question 2

Part a: Calculate the specific heat capacity of water using the given data.

Model Answer: Using $Q = mc\Delta t$: 67,990 J = (250.0 g)(c)(85.0°C - 20.0°C) 67,990 J = (250.0 g)(c)(65.0°C) c = 67,990 J ÷ (250.0 g × 65.0°C) c = 67,990 J ÷ 16,250 g°C c = 4.184 J/g°C (2 marks) (2 marks)

Part b: Explain why water has a relatively high specific heat capacity compared to most substances.

Model Answer: Water has a high specific heat capacity because water molecules form hydrogen bonds with each other. These hydrogen bonds require additional energy to overcome during heating. The energy goes into increasing molecular motion (vibrational and rotational) without breaking the hydrogen bonds, which would cause a phase change. This makes water an excellent thermal reservoir. (2 marks) (2 marks)

Part c: State what happens to the hydrogen bonds when water is heated within the liquid phase.

Model Answer: The hydrogen bonds remain intact while the molecular motion increases. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for correct setup of $Q = mc\Delta t$; 1 mark for correct calculation of c = 4.184 J/g°C

Part b (2 marks): 1 mark for mentioning hydrogen bonds; 1 mark for explaining energy requirement

Part c (1 mark): 1 mark for stating bonds remain intact during liquid phase heating

Question 3

Part a: Explain how the second law of thermodynamics relates to spontaneous heat flow.

Model Answer: The second law states that entropy (disorder) of the universe must increase. Heat flows spontaneously from hot to cold because this increases total entropy. When fast-moving molecules transfer energy to slow-moving molecules, the energy becomes more evenly distributed throughout the system, increasing overall disorder and making the process thermodynamically favored. (2 marks) (2 marks)

Part b: Describe what thermal equilibrium means in terms of molecular kinetic energy.

Model Answer: At thermal equilibrium, all molecules in the system have the same average kinetic energy. Since temperature represents average kinetic energy, this means both substances reach the same temperature. Energy transfer continues until this state is achieved, where molecular collisions no longer result in net energy transfer. (2 marks) (2 marks)

Part c: State the fundamental principle that allows us to write $Q_{lost} = Q_{gained}$ in calorimetry.

Model Answer: Energy conservation - energy cannot be created or destroyed, only transferred. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for stating entropy must increase; 1 mark for explaining energy distribution

Part b (2 marks): 1 mark for same average KE; 1 mark for relating to temperature

Part c (1 mark): 1 mark for energy conservation principle

Question 4

Part a: Calculate the heat gained by the water in this experiment.

Model Answer: $Q_{water} = mc\Delta t = (200.0\ g)(4.184\ J/g°C)(28.5°C - 22.0°C)$ $Q_{water} = (200.0\ g)(4.184\ J/g°C)(6.5°C)$ $Q_{water} = 5439.2\ J$ (2 marks) (2 marks)

Part b: Calculate the specific heat capacity of the metal using energy conservation principles.

Model Answer: $Q_{metal} = -Q_{water}$ (energy conservation) $(50.0\ g)(c_{metal})(28.5°C - 95.0°C) = -5439.2\ J$ $(50.0\ g)(c_{metal})(-66.5°C) = -5439.2\ J$ $(3325\ g°C)(c_{metal}) = 5439.2\ J$ $c_{metal} = 1.64\ J/g°C$ (2 marks) (2 marks)

Part c: State which industry uses calorimetry to determine specific heat capacities of materials like this metal.

Model Answer: The metallurgical industry. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for correct formula; 1 mark for correct calculation = 5439.2 J

Part b (2 marks): 1 mark for using $Q_{metal} = -Q_{water}$; 1 mark for correct $c_{metal} = 1.64\ J/g°C$

Part c (1 mark): 1 mark for metallurgical industry

Question 5

Part a: Explain the energy transformation pathway from solar energy to fossil fuels.

Model Answer: Solar photons are captured through photosynthesis, which forms chemical bonds storing energy. Through geological transformation over time, biological matter is converted to fossil fuels. When fossil fuels undergo combustion, they release exactly the amount of energy originally captured from sunlight, regardless of the geological time scales or complex biochemical pathways involved. (2 marks) (2 marks)

Part b: Explain why hydrocarbons have higher energy density than alcohols based on bond structure.

Model Answer: The energy content of molecules depends on their bond structure. C-H bonds store more energy than C-O bonds. Since hydrocarbons contain primarily C-H bonds while alcohols contain C-O bonds (in addition to C-H bonds), hydrocarbons have higher energy density. This explains why removing oxygen atoms from biological matter creates more energy-dense fossil fuels. (2 marks) (2 marks)

Part c: State what type of calorimeter food scientists use to measure caloric content.

Model Answer: Bomb calorimeter. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for photosynthesis and bond formation; 1 mark for geological transformation and energy conservation

Part b (2 marks): 1 mark for C-H vs C-O bond energy; 1 mark for relating to energy density

Part c (1 mark): 1 mark for bomb calorimeter

Question 6

Part a: Explain what aspects of a chemical reaction catalysts can and cannot change.

Model Answer: Catalysts can affect reaction rates by lowering activation energy, making reactions proceed faster. However, catalysts never change enthalpy changes (ΔH) or equilibrium positions. The total energy released or absorbed by a reaction remains the same regardless of catalyst presence, and the equilibrium concentrations of products and reactants are unchanged. (2 marks) (2 marks)

Part b: Explain the difference between activation energy and enthalpy change in determining reaction characteristics.

Model Answer: Activation energy affects reaction rates and initiation requirements - it determines how fast a reaction proceeds and whether external energy input is needed to start. Enthalpy change determines the total energy released or absorbed and the thermodynamic favorability of the reaction. A reaction can be thermodynamically favorable (negative ΔH) but still require high activation energy to proceed. (2 marks) (2 marks)

Part c: State why even highly exothermic reactions may require external energy input.

Model Answer: They need to overcome the activation energy barrier to initiate the reaction. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for what catalysts change (rates); 1 mark for what they don't change (ΔH, equilibrium)

Part b (2 marks): 1 mark for activation energy effects; 1 mark for enthalpy change effects

Part c (1 mark): 1 mark for activation energy requirement

Question 7

Part a: List the four components that must be included in an energy analysis of this industrial process.

Model Answer: The analysis must include: (1) predicted enthalpy change using formation data, (2) identification of whether external heating will be required, (3) catalyst requirements based on activation energy, and (4) safety considerations based on energy release calculations. (2 marks) (2 marks)

Part b: Explain how thermochemical calculations provide the foundation for industrial applications.

Model Answer: Thermochemical calculations enable quantitative energy predictions through standardized formation enthalpies and Hess's law. These predictions provide the foundation for industrial process design by determining energy requirements, and for fuel efficiency analysis by calculating energy output. This allows optimization of chemical processes and proper energy planning. (2 marks) (2 marks)

Part c: State what principle allows enthalpy calculations to be independent of reaction pathway.

Model Answer: Hess's law (or energy conservation). (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 0.5 marks for each of the four required components

Part b (2 marks): 1 mark for quantitative predictions; 1 mark for industrial applications

Part c (1 mark): 1 mark for Hess's law or energy conservation

Question 8

Part a: Show the complete calculation that confirms this energy requirement using $Q = mc\Delta t$.

Model Answer: $Q = mc\Delta t = (250.0\text{ g})(4.184\text{ J/g°C})(65.0\text{°C})$ $Q = 67{,}990\text{ J}$ Converting to kJ: $67{,}990\text{ J} \div 1000 = 68.0\text{ kJ}$ The observation is confirmed. (2 marks) (2 marks)

Part b: Explain what happens to water molecules when this energy is absorbed without phase change.

Model Answer: The absorbed energy increases the vibrational and rotational motion of water molecules within the liquid phase. This increases the average kinetic energy of the molecules, causing the temperature to rise. The hydrogen bonds between water molecules remain intact - they are not broken as that would cause a phase change to steam. (2 marks) (2 marks)

Part c: State what each variable in the equation $Q = mc\Delta t$ represents with its units.

Model Answer: Q = heat energy (joules), m = mass (grams), c = specific heat capacity (J/g°C), Δt = temperature change (°C). (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for correct calculation setup; 1 mark for conversion and confirmation

Part b (2 marks): 1 mark for increased molecular motion; 1 mark for hydrogen bonds intact

Part c (1 mark): 0.25 marks for each correctly identified variable with units

Question 9

Part a: Explain the relationship between bond types and energy content in organic molecules.

Model Answer: The energy content depends directly on bond structure. C-H bonds store more energy than C-O bonds. This explains why hydrocarbons (with many C-H bonds) have higher energy density than alcohols (which contain C-O bonds). More C-H bonds mean higher energy content per unit mass. (2 marks) (2 marks)

Part b: Explain how geological processes create energy-dense fossil fuels from biological matter.

Model Answer: Geological processes remove oxygen atoms from biological matter over long time periods. Since C-O bonds store less energy than C-H bonds, removing oxygen increases the proportion of high-energy C-H bonds. This transformation creates more energy-dense fossil fuels from the original biological material that captured solar energy through photosynthesis. (2 marks) (2 marks)

Part c: State why physical states must be specified in thermochemical equations.

Model Answer: Intermolecular forces affect phase-dependent enthalpy values. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for C-H vs C-O bond energy; 1 mark for relating to energy density

Part b (2 marks): 1 mark for oxygen removal; 1 mark for increased energy density

Part c (1 mark): 1 mark for intermolecular forces affecting enthalpy

Question 10

Part a: Explain how calorimetry bridges theoretical calculations with experimental reality.

Model Answer: Calorimetry uses the $Q = mc\Delta t$ equation to measure actual energy changes in experiments. These measurements validate predicted thermochemical values from theoretical calculations. Calorimetry also enables determination of unknown thermochemical parameters that cannot be calculated theoretically, providing real-world data to confirm or refine theoretical models. (2 marks) (2 marks)

Part b: Explain why the equation $Q_{lost} = Q_{gained}$ can be used in calorimetry calculations.

Model Answer: This equation is based on energy conservation - energy cannot be created or destroyed, only transferred from one substance to another. In an isolated system (like a calorimeter), the total energy remains constant. Therefore, the heat lost by the hot object must exactly equal the heat gained by the cold object. (2 marks) (2 marks)

Part c: State three industries mentioned that use calorimetry applications.

Model Answer: Beverage companies, metallurgical industries, and food science. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for validation of predictions; 1 mark for determining unknowns

Part b (2 marks): 1 mark for energy conservation; 1 mark for isolated system explanation

Part c (1 mark): 1 mark for correctly listing three industries

Question 11

Part a: Explain three key distinctions that diploma exams frequently test according to the lesson.

Model Answer: Diploma exams frequently test: (1) the distinction between activation energy (affects rates) and enthalpy change (determines energy released/absorbed), (2) proper scaling of ΔH values with equation coefficients when reactions are multiplied, and (3) correct application of Hess's law to calculate enthalpy changes for reactions. (2 marks) (2 marks)

Part b: Explain why enthalpy changes are independent of reaction pathway while activation energy is not.

Model Answer: Enthalpy change depends only on initial and final states due to energy conservation (Hess's law), making it pathway-independent. The same amount of energy is released/absorbed regardless of how the reaction proceeds. Activation energy, however, represents the energy barrier that must be overcome, which can vary with different pathways or catalysts that provide alternative routes. (2 marks) (2 marks)

Part c: State what students must understand about catalysts' effects on equilibrium.

Model Answer: Catalysts never change equilibrium positions. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 0.67 marks for each of three distinctions

Part b (2 marks): 1 mark for enthalpy pathway independence; 1 mark for activation energy variation

Part c (1 mark): 1 mark for no effect on equilibrium

Question 12

Part a: Explain this observation using the concepts of temperature and heat.

Model Answer: Temperature represents average kinetic energy of molecules, while heat represents total kinetic energy transfer. A large mass of lukewarm water has many molecules, each with moderate kinetic energy. A small mass of hot water has few molecules with high kinetic energy. The total energy (heat) can be greater in the large mass despite lower average energy (temperature) per molecule. (2 marks) (2 marks)

Part b: Predict the direction of heat flow if these two water samples are brought into contact.

Model Answer: Heat will flow from the hot water (higher temperature/average kinetic energy) to the lukewarm water (lower temperature), regardless of which contains more total heat. Heat flow depends on temperature difference, not total heat content. Flow continues until both reach the same temperature (thermal equilibrium). (2 marks) (2 marks)

Part c: State what determines the direction of spontaneous heat flow.

Model Answer: Temperature difference (heat flows from higher to lower temperature). (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for distinguishing temperature vs heat; 1 mark for explaining total vs average

Part b (2 marks): 1 mark for correct direction; 1 mark for temperature-based explanation

Part c (1 mark): 1 mark for temperature difference

Question 13

Part a: List the complete cause-effect sequence from solar photons to optimized chemical processes.

Model Answer: Solar photons → photosynthesis → chemical bond formation → geological transformation → fossil fuel creation → combustion → heat release → calorimetric measurement → enthalpy determination → thermochemical calculations → industrial energy planning → optimized chemical processes. (2 marks) (2 marks)

Part b: Explain how energy conservation applies throughout this entire transformation sequence.

Model Answer: Energy is conserved at every step - it cannot be created or destroyed, only transformed. The energy in fossil fuels equals the solar energy originally captured through photosynthesis. Combustion releases exactly this stored amount. Energy conservation allows us to track and calculate energy through all transformations, regardless of time scales or pathway complexity. (2 marks) (2 marks)

Part c: State what process captures solar energy and stores it in chemical bonds.

Model Answer: Photosynthesis. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 2 marks for complete sequence (partial credit for mostly correct)

Part b (2 marks): 1 mark for conservation principle; 1 mark for application throughout sequence

Part c (1 mark): 1 mark for photosynthesis

Question 14

Part a: Describe the experimental setup needed to determine the metal's specific heat capacity.

Model Answer: Place a known mass of hot metal at a measured temperature into a known mass of water at a measured temperature in an insulated container (calorimeter). Measure the final equilibrium temperature. Use the principle that heat lost by metal equals heat gained by water, applying $Q = mc\Delta t$ to both substances to solve for the unknown specific heat capacity. (2 marks) (2 marks)

Part b: Explain why this calculation requires the assumption of an isolated system.

Model Answer: In an isolated system, no energy enters or leaves, ensuring total energy remains constant (energy conservation). This allows us to equate heat lost by the metal to heat gained by water ($Q_{metal} = -Q_{water}$). If the system weren't isolated, energy could escape to surroundings or enter from outside, invalidating the calculation. (2 marks) (2 marks)

Part c: Calculate the product (3325 g°C) if solving for a metal's specific heat when heat transfer equals 5439.2 J.

Model Answer: c_{metal} = 5439.2 J ÷ 3325 g°C = 1.64 J/g°C (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for setup description; 1 mark for calculation method

Part b (2 marks): 1 mark for energy conservation; 1 mark for isolation necessity

Part c (1 mark): 1 mark for correct calculation

Question 15

Part a: Explain what molecular feature gives water its high specific heat capacity.

Model Answer: Water molecules form hydrogen bonds with each other. These intermolecular hydrogen bonds require additional energy to overcome during heating. The energy input goes into increasing molecular motion (vibrational and rotational) rather than breaking bonds, requiring more energy per degree of temperature change compared to substances without hydrogen bonding. (2 marks) (2 marks)

Part b: Explain why water is described as an excellent thermal reservoir.

Model Answer: Water's high specific heat capacity (4.184 J/g°C) means it can absorb or release large amounts of heat with relatively small temperature changes. This property allows water to store thermal energy effectively and moderate temperature fluctuations, making it ideal for cooling systems, climate regulation, and energy storage applications. (2 marks) (2 marks)

Part c: State the specific heat capacity value for water used in calculations.

Model Answer: 4.184 J/g°C (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for hydrogen bonding; 1 mark for energy requirement explanation

Part b (2 marks): 1 mark for heat capacity effect; 1 mark for practical applications

Part c (1 mark): 1 mark for 4.184 J/g°C

Question 16

Part a: Define entropy and explain its role in spontaneous heat transfer.

Model Answer: Entropy is a measure of disorder or randomness in a system. The second law states that entropy of the universe must increase. Spontaneous heat transfer from hot to cold increases entropy because energy becomes more evenly distributed among molecules, creating greater disorder. This makes the process thermodynamically favored. (2 marks) (2 marks)

Part b: Describe what happens to the entropy when fast-moving molecules transfer energy to slow-moving molecules.

Model Answer: When fast molecules transfer energy to slow molecules, the energy distribution becomes more uniform. Instead of having distinct populations of high-energy and low-energy molecules, the system moves toward a more even energy distribution. This increased randomness in energy distribution represents higher entropy, making the process spontaneous. (2 marks) (2 marks)

Part c: State which law of thermodynamics requires entropy to increase.

Model Answer: The second law of thermodynamics. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for entropy definition; 1 mark for role in heat transfer

Part b (2 marks): 1 mark for energy distribution; 1 mark for entropy increase

Part c (1 mark): 1 mark for second law

Question 17

Part a: Explain how formation enthalpies and Hess's law enable industrial energy planning.

Model Answer: Standardized formation enthalpies provide reference values for calculating energy changes in any reaction. Hess's law allows calculation of enthalpy changes for complex reactions by combining simpler reactions, since enthalpy is pathway-independent. Together, they enable quantitative predictions of energy requirements or releases, allowing industries to plan heating/cooling needs and optimize processes. (2 marks) (2 marks)

Part b: Explain why understanding both activation energy and enthalpy change is crucial for industrial processes.

Model Answer: Activation energy determines reaction rates and whether external energy input is needed to initiate reactions - crucial for process timing and startup costs. Enthalpy change determines total energy released or absorbed and thermodynamic favorability - essential for energy balance and safety. Both must be considered: a thermodynamically favorable reaction may still need heating to overcome activation barriers. (2 marks) (2 marks)

Part c: State what safety consideration is based on energy release calculations.

Model Answer: Heat management/thermal hazards from exothermic reactions. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for formation enthalpies; 1 mark for Hess's law application

Part b (2 marks): 1 mark for activation energy role; 1 mark for enthalpy change role

Part c (1 mark): 1 mark for heat/thermal safety

Question 18

Part a: Calculate the temperature change of the metal and explain why it is negative.

Model Answer: Δt = final temperature - initial temperature = 28.5°C - 95.0°C = -66.5°C The temperature change is negative because the metal cooled down (lost heat). The final temperature is lower than the initial temperature, and by convention, Δt is always calculated as final minus initial. (2 marks) (2 marks)

Part b: Set up the equation to solve for the metal's specific heat capacity using the given information.

Model Answer: Since $Q_{metal} = -Q_{water}$ and Q_{water} = 5439.2 J: Q_{metal} = -5439.2 J Using $Q = mc\Delta t$: -5439.2 J = (50.0 g)(c_{metal})(-66.5°C) 5439.2 J = (3325 g°C)(c_{metal}) c_{metal} = 1.64 J/g°C (2 marks) (2 marks)

Part c: State why the heat lost by metal equals the heat gained by water in magnitude.

Model Answer: Energy conservation in an isolated system. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for calculation; 1 mark for explanation of negative value

Part b (2 marks): 1 mark for correct setup; 1 mark for solving for c_{metal}

Part c (1 mark): 1 mark for energy conservation

Question 19

Part a: Explain how the energy in fossil fuels relates to the original solar energy captured.

Model Answer: All chemical energy in fossil fuels originated from solar energy captured through photosynthesis by ancient organisms. This energy was stored in chemical bonds of organic molecules. Through geological processes over millions of years, this biological matter transformed into fossil fuels. Energy conservation ensures that combustion releases exactly the amount of energy originally captured from sunlight. (2 marks) (2 marks)

Part b: Explain why removing oxygen atoms during fossilization increases energy density.

Model Answer: C-H bonds store more energy than C-O bonds. When geological processes remove oxygen atoms from biological matter, they eliminate lower-energy C-O bonds while preserving or concentrating higher-energy C-H bonds. This increases the energy per unit mass, creating more energy-dense fossil fuels from the original biological material. (2 marks) (2 marks)

Part c: State the biological process that initially captures solar energy.

Model Answer: Photosynthesis. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for photosynthesis origin; 1 mark for energy conservation

Part b (2 marks): 1 mark for bond energy differences; 1 mark for concentration effect

Part c (1 mark): 1 mark for photosynthesis

Question 20

Part a: Identify four key measurement or calculation steps in the energy transformation sequence after combustion occurs.

Model Answer: After combustion, the sequence includes: (1) heat release measurement, (2) calorimetric measurement using $Q = mc\Delta t$, (3) enthalpy determination from experimental data, and (4) thermochemical calculations using formation data and Hess's law. These steps bridge experimental measurements with theoretical predictions for industrial planning. (2 marks) (2 marks)

Part b: Explain how calorimetry validates theoretical thermochemical predictions.

Model Answer: Calorimetry provides experimental measurements of actual energy changes using $Q = mc\Delta t$. These measured values can be compared with theoretical predictions from formation enthalpies and Hess's law calculations. Agreement between experimental and theoretical values validates the predictions, while discrepancies reveal factors not accounted for in theory, enabling refinement of models. (2 marks) (2 marks)

Part c: State the final outcome of the complete cause-effect energy transformation chain.

Model Answer: Optimized chemical processes. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 0.5 marks for each of four steps identified

Part b (2 marks): 1 mark for experimental validation; 1 mark for comparison/refinement

Part c (1 mark): 1 mark for optimized chemical processes